



Systems neuroendocrinology in ME/CFS and long COVID: a chronobiological framework for hormone-based research

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ABSTRACT

Hormonal dysregulation is increasingly reported in ME/CFS and Long COVID, yet the broader role of neuroendocrine disruption in these conditions remains underexplored. While changes in steroid, peptide, and neuropeptide hormones have been identified, these findings are often considered in isolation and without attention to their timing or integration within broader physiological systems. The hypothalamic-pituitary axes regulate endocrine, immune, autonomic, nervous, and metabolic functions, systems commonly affected in both conditions, yet their circadian and menstrual dynamics are rarely investigated.

In this review, we examine the evidence for neuroendocrine dysfunction in ME/CFS and Long COVID, focusing on hormone output, functional assays, receptor expression, and the coordination of endocrine biorhythms. Sex hormone signalling emerges as a key area of vulnerability, particularly given the female predominance in both conditions and the complexity of reproductive hormone regulation.

We argue that accurate hormone measurement and time-structured sampling, including circadian and menstrual rhythms, are essential for detecting meaningful biological differences. By embedding chronobiology-aware, dense-sampling strategies and integrating multi-omic analyses into multi-system study designs, we outline a framework for investigating dynamic endocrine mechanisms underlying symptom variability and multisystem dysfunction, which may ultimately support the development of more targeted, personalised interventions.

1. Introduction

1.1. Myalgic encephalomyelitis/chronic fatigue syndrome and long COVID

Myalgic Encephalomyelitis/Chronic Fatigue Syndrome (ME/CFS) and Post-Acute Sequelae of Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) Infection 'Long COVID', are debilitating, multifaceted conditions that face significant challenges in diagnosis, treatment, policy, and research due to their complexity and widespread misunderstanding (Davis et al., 2023). Neither illness has a definitive

diagnostic test nor established treatments. For many, these illnesses are long-term, severely disabling, and bear poor prognosis, with less than 10% of patients achieving full recovery (Davis et al., 2023; Ghali et al., 2022; Friedberg et al., 2020).

ME/CFS is characterised by a profound fatigue considered pathological, with patients experiencing a substantial reduction in activity levels that persist for at least 6 months. The ME/CFS diagnosis of 'last resort' can be a lengthy and challenging process, often taking several years due to the absence of specific diagnostic tests, and the need to rule out other medical conditions with similar symptoms (Bested and Marshall, 2015; Fukuda et al., 1994). As defined by the Canadian Consensus Criteria (CCC) (Carruthers et al., 2011), ME/CFS patients

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